

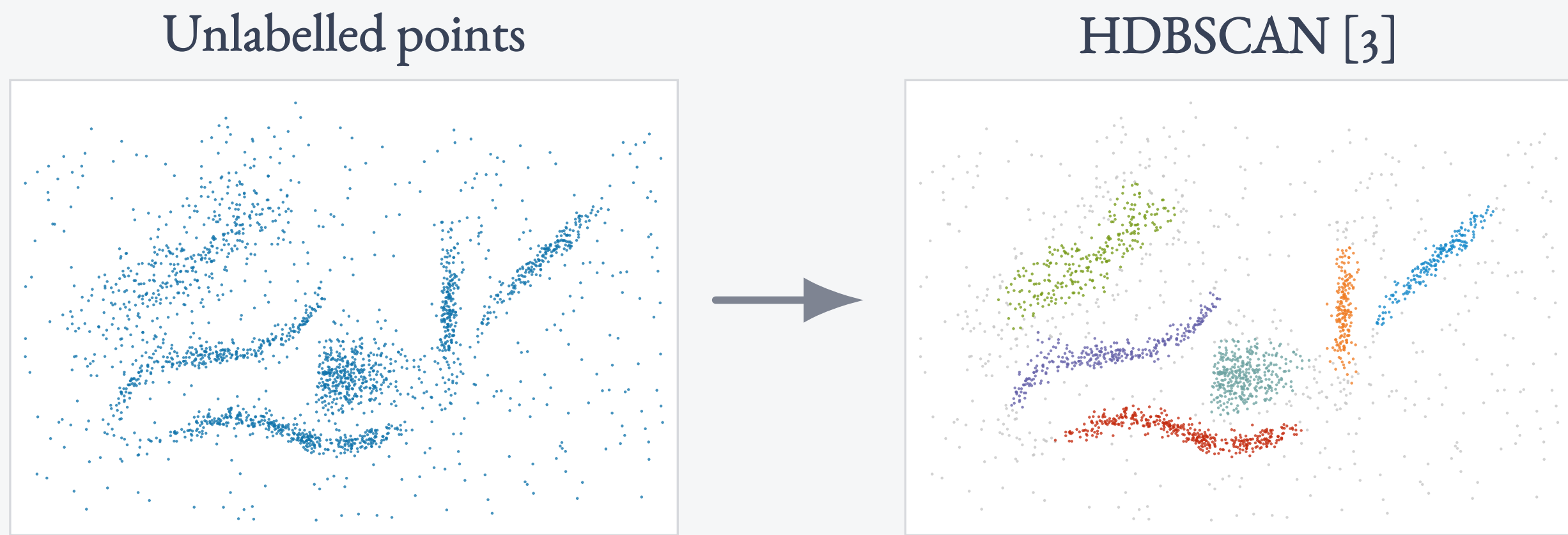
Higher-order clustering for 3D point clouds

Exact density hierarchies and geometric tokens for 3D learning

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1 Clustering

Find groups without class labels. Recover non-convex shapes and leave noisy points unassigned.



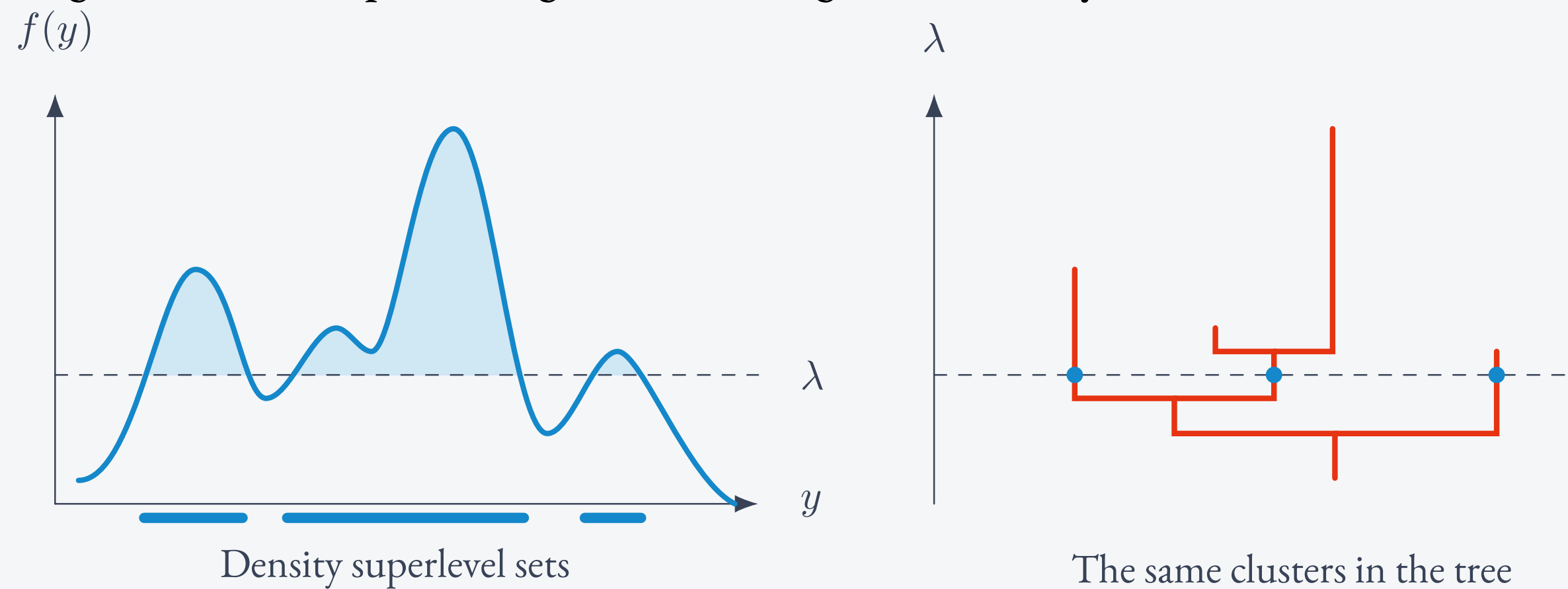
Colours: clusters; grey: unassigned points [3].

2 HARTIGAN's model and hierarchy

Clusters are connected components of density superlevel sets [1].

$$E_\lambda = \{y \in \mathbb{R}^d : f(y) \geq \lambda\}$$

Lowering λ makes components grow and merge: a hierarchy.



Key quantity: the neighbour radius $r_K(y)$.

$$r_K(y) = \min\{r \geq 0 : \#(X \cap B(y, r)) \geq K\}$$

Distance from y to its K -th nearest sample point [2].

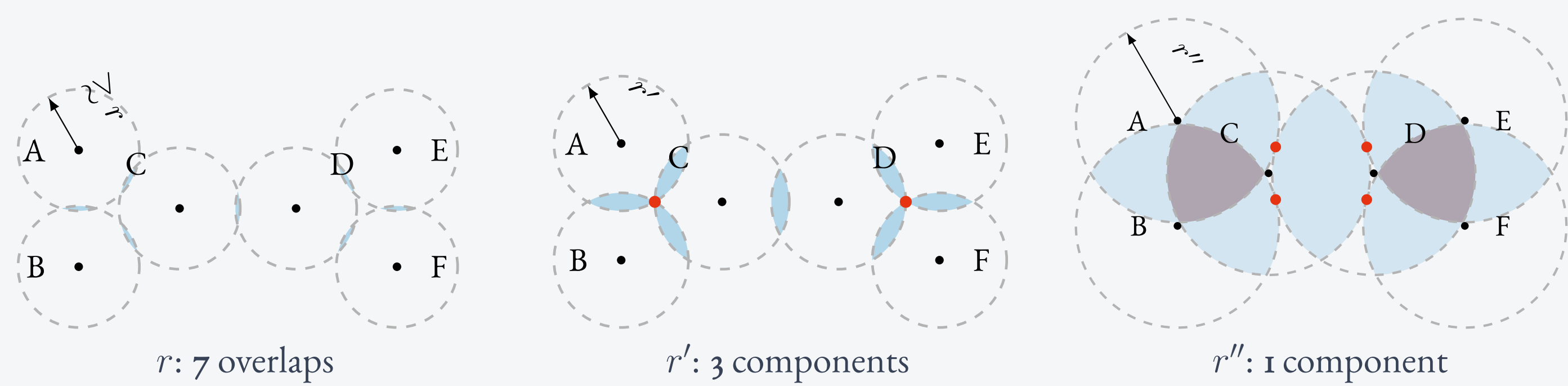
$$\hat{f}_K(y) = \frac{K}{n \omega_d r_K(y)^d}$$

n : number of points; ω_d : unit-ball volume.

3 Same six points, different hierarchies

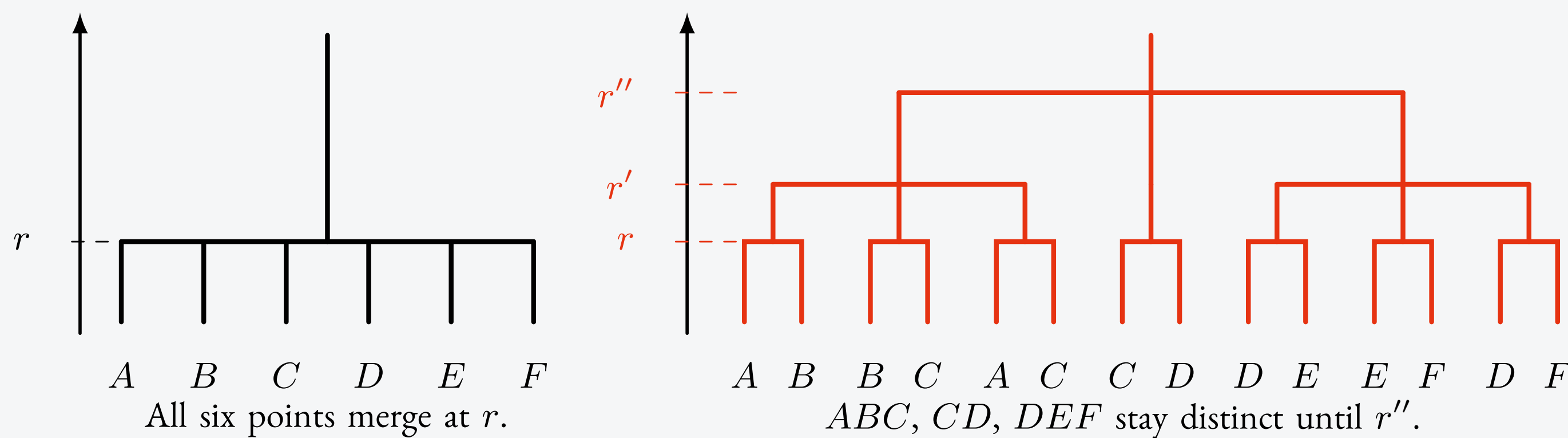
For $K = 2$, track the region covered by at least two balls:

$$E_{2,\rho} = \{y : r_2(y) \leq \rho\} = \bigcup_{\{x_i, x_j\} \subset X} (B(x_i, \rho) \cap B(x_j, \rho))$$



DBSCAN / Robust SL

2-NN HARTIGAN



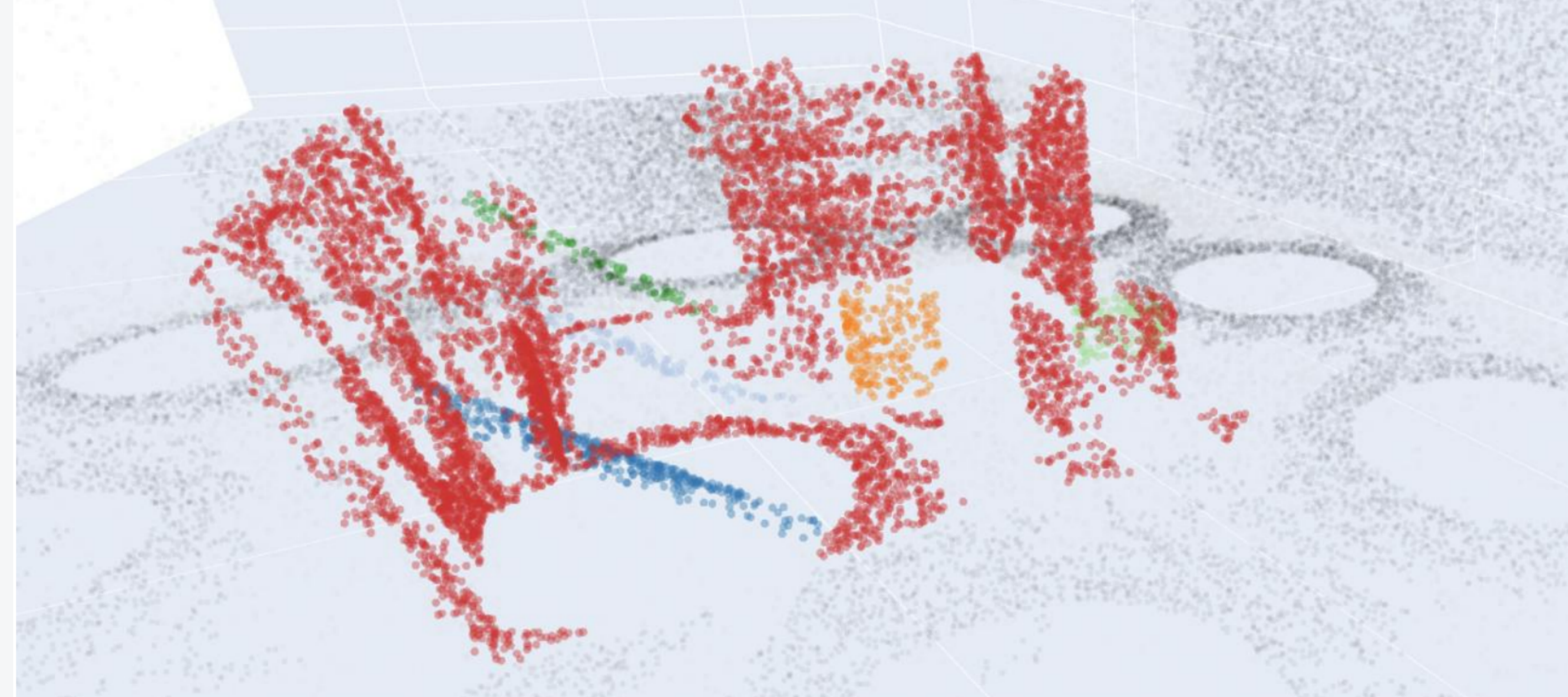
HGP-Clusterer connects edges through triangles, recovering the exact hierarchy [4,5]. Its branches may share points.

7 References

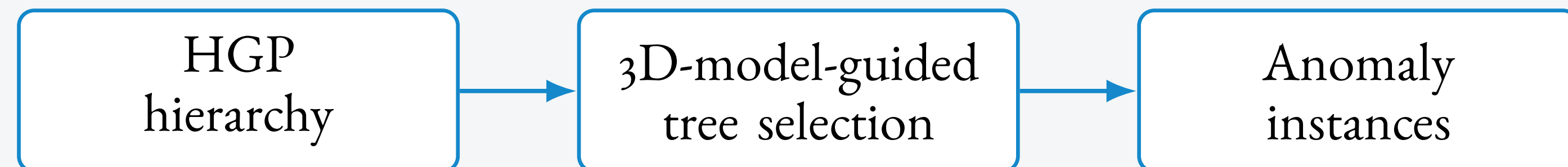
- [1] J. A. Hartigan. *Consistency of single linkage for high-density clusters*. JASA, 76:388–394, 1981.
- [2] G. Biau and L. Devroye. *Lectures on the Nearest Neighbor Method*. Springer, 2015.
- [3] R. J. G. B. Campello, D. Moulavi and J. Sander. *Density-Based Clustering Based on Hierarchical Density Estimates*. PAKDD, pp. 160–172, 2013.

4 LiDAR: detect unexpected objects

Objects present in a scan, absent from a reference 3D model. With Marie ASPRO (Inria Startup Studio).



Synthetic benchmark: 2 million points. Red: model-consistent; other colours: anomalies; grey: background. © Naval Group [6].



Geometric priors select branches. No training required.

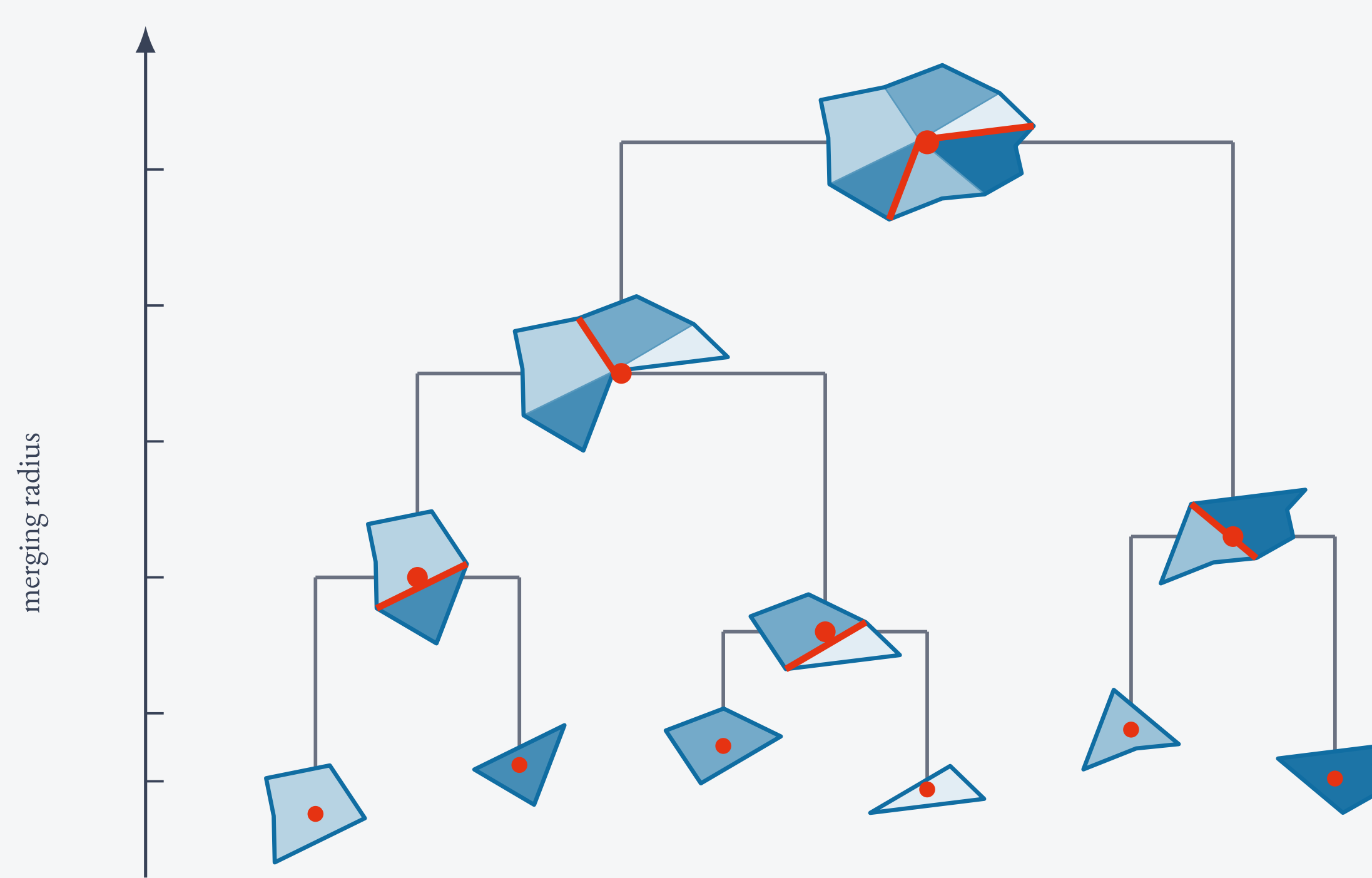
5 Perspective: geometry-guided 3D learning

The same object, different sampling.

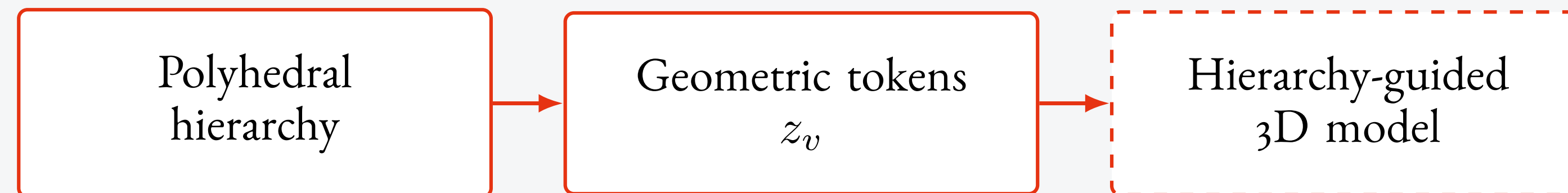


Range, scan pattern and occlusion change the point cloud. Surface geometry may be more stable.

Polyhedra as tokens; hierarchy as multiscale context.



Each parent joins its children along shared faces (red). Schematic geometry and merging radii.



Research hypothesis: guide a 3D foundation model with geometric tokens and their merging scales. Learning gains remain to be tested.

6 Inria Startup Studio: Percolia



Louis HAUSEUX & Alban HAUSEUX
Entering Inria Startup Studio.
Geometry-aware tools for 3D perception.



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